

Optimized Pipelined Carry-Save Adder (PCSA) for 16-Bit Cryptographic Arithmetic Units

Abstract— The rapid and energy-efficient arithmetic operations are a necessity in cryptographic protocols wherein the speed of computation has a direct relationship with encryption and decryption speed on a real-time basis. The current work focuses on optimizing a 16-bit pipelined Carry-Save Adder (CSA) architecture designed for use in cryptographic arithmetic operations, implemented and evaluated on a Xilinx Artix-7 FPGA. The suggested design follows multiple stage pipelined method to parallelize the process of computing partial sums and carries thereby significantly reducing critical path delay and enhancing overall throughput. An in-depth trade-off comparison of the Ripple Carry Adder (RCA), the standard CSA as well as the proposed one, the Pipelined CSA shows significant performance improvement. Simple but with high propagation delay 12.8 ns the RCA is a sequential carry generator. Partial carry parallelization provides moderate improvement 7.2 ns with the conventional CSA. By contrast, the proposed pipelined CSA has a delay of 3.5 ns, the upper limit of the operating frequency of 100 MHz and a 25 percent power reduction, whereas the balanced resource utilization of the Artix-7 FPGA is achieved. This performance has shown that the pipelined architecture is efficient in real time cryptography applications, such as key creation, modular arithmetic and encryption cores, that demand both speed and energy meeting.

Keywords— *Carry-Save Adder, Ripple Carry Adder, Pipelined Architecture, FPGA, Cryptographic Arithmetic Unit, Xilinx Artix-7*

Introduction

Designing high-speed and energy-efficient adders remains a major challenge in modern cryptographic hardware development. Traditional architectures including Ripple Carry and Carry Save Adders are usually limited in propagation delay and throughput particularly in arithmetic intensive cryptographic tasks. In order to address these issues, this piece of work presents an optimized 16-bit pipelined Carry-Save Adder (CSA) architecture to speed up and efficiently improve cryptographic arithmetic devices. Integrating pipeline registers across computation stages significantly reduces critical path delay and increases operating frequency, enabling real-time processing of cryptographic data streams. The presented design has been placed on a Xilinx Artix-7 FPGA, which illustrates the equilibrium between the area, power and delay, thus confirming the appropriateness of the design in high-performance and low-power cryptographic functions.

I. LITERATURE SURVEY

In this paper, I present a carry-save adder (CSA) array that is both energy- and area-efficient that has been customized to carry out spiking transformer multiplication of matrices. Its design uses a two-stage CSA structure, which allows reusing of logic between 1-bit self-attention and 8-bit linear mode, and a cubic-mesh structure to ensure reuse of weight across timesteps is as high as possible. The proposed architecture can save 2.7x area and 3.75x the power of the traditional accumulation designs, and this indicates that it has the good prospect of integration in digital neuromorphic accelerators [1]. This paper is about a Hybrid Carry-Save Adder (CSA) architecture that can help enhance speed and area efficiency of addition in architecture 32-bit. Reduced final addition bits in the LSB of the product take place by moving the carry towards, the LSB of the partial products thus enabling piping of the implanting process and improve performance. The proposed method also does not involve the accumulation phase which most of the time causes the delay and it combines it with the compression of the partial products. The design is coded in VHDL and synthesized through Xilinx tools and it shows optimized hardware performance [2].

This work is based on the design and analysis of a Carry-Save Adder (CSA) which is a type of combinational logic circuit that is employed to add three or more multi-bit binary numbers efficiently. The CSA produces parallel outputs which consist of partial sum bits and carry bits, unlike more traditional adders, which has the minimal propagation delay and enables high clock rates. The work and tests CSA with the help of Tanner EDA, Xilinx, and Multisim tools in comparison to existent carry-skip adders. In the findings, the suggested CSA architecture is capable of performing the computations more quickly and requiring smaller circuit models when implementing digital systems in the future [3]. In this paper, the author is presenting an adapted version of a Carry-Save Adder (CSA) architecture that is designed to run high speed and low area efficient computing both in digital and signal processing systems. The design uses a multiplexer-based full adder to assemble all CSA stages, which has the effect of minimizing the circuit complexity, area and delay. The proposed 64-bit adder was implemented in VHDL and synthesized on an Altera Cyclone II FPGA with Quartus II, achieving a 17.75% improvement in delay, 1.60% improvement in power use, and also an 8.81% improvement in the logic use over the current CSA designs proving that the proposed 64-bit adder is suitable in high-performance systems [4].

It has been pointed out in the past that as a component is a digital integrated circuit, adders are necessities with minimal delay, low power consumption, and area minimality being the

most important design considerations. There have been comparative studies of several different adder designs such as Ripple Carry, Carry Look-Ahead, Carry Skip, Carry Select, Carry Increment, Carry Save, and Carry Bypass Adders based on 0.12 μ m CMOS technology in Micro wind. The findings provide a demonstration of the trade-offs inherent in delay, power, and area that can be used to select the effective adder topologies used in high-performance arithmetic circuits [5].

The analysis presents an Error-Tolerant Adder (ETA) which utilizes the principle of error tolerance in VLSI design to loosen the accuracy requirements and achieve a huge improvement in speed and power consumption. Experiments indicate an upward response of more than 65 percent in Power-Delay Product (PDP) over traditional adders. The applications that are particularly masterful of such architecture are in digital signal processing systems in which small errors in calculation are acceptable in order to achieve high performance and low energy consumption [6]. The research is dedicated to maximizing of power dissipation, an important consideration of design of integrated circuits after speed. Adders are very important in most digital systems; therefore, the paper is a comparative study on different high speed adder architectures in terms of area, delay, and power consumption. A modified Carry Select Adder (CSLA) is suggested and evaluated at a series of phases and proves to be more efficient in terms of area consumption and power saving than the traditional designs [7].

This paper discusses the issues of power and performance in the contemporary portable VLSI systems through the proposal of a Modified Area-Efficient Carry Select Adder (MA-CSLA). The design incorporates a Binary to Excess-1 Converter (BEC) in it to minimize the number of gates and complexity of hardware technologies. Experimental results of 8-, 16-, 32-, and 64-bit SQRT CSLA architectures show that the modified implementation realizes dramatic savings both in area and power consumption and similar delay performance. Therefore, the MA-CSLA is a good choice in low-power (VLSI) applications that need speed and power efficiency (low power) and is also able to utilize silicon (low power). The Carry Select Adder (CSLA) has been known to perform high arithmetic speed in calculating machines. To be more efficient a modified CSLA architecture was created where the Binary to Excess-1 Converter (BEC) was substituted with Common Boolean Logic (CBL) strategy. This modification is favorable in logic design since it leads to less area consumption and power consumption in addition to faster computing process over traditional ways of implementing CSLA [9].

In this work, a comparative analysis of a 32-bit array multiplier built on both Carry Look-Ahead (CLA) and Carry Save Adder (CSA) implementation and modeled in VHDL is presented. Key measures of performance, which are area, speed, and power consumption, are evaluated in the study. Experiments have shown that CSA-based multiplier has much better computational speed with about 78 percent better speed, 4 percent less area, and 1 percent less power consumption than the CLA-based design. This shows how CSA structures are applicable to high speed, power efficient arithmetic units of modern VLSI systems [10].

II. PRELIMINARIES

A. 16-bit Ripple Carry Adder (RCA)

It is constructed as a 16-bit ripple carry adder (RCA) constructed by tapering 16 full adders, each adding twice as many bits with a carry-in, giving the result known as a sum bit, and a carry-out known as the carry-out. The critical feature of this design is that the carry output of each full adder ripples through to the next stage, leading to the total addition delay being proportional to the number of bits, which in this case is 16.

Despite its simplicity and ease of implementation on FPGA, the main drawback of the RCA is its relatively slow speed due to carry propagation delay, which can significantly impact performance in high-speed applications. Its delay is the sum of the propagation delay of each full adder in the chain, which makes it less suitable for large-bit-width operations where speed is crucial.

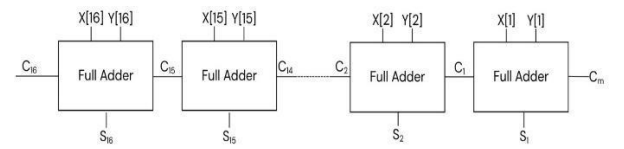


Fig.1 Block diagram of RCA

B. Drawbacks

- **Slow Performance:** The carry signal propagates sequentially through all bit positions, causing a delay that scales linearly with the adder's bit width.
- **Not Suitable for High-Speed Applications:** The propagation delay limits efficiency in large bit-width adders, such as 32-bit or 64-bit implementations.
- **Sequential Operation:** Since each full adder relies on the carry from the preceding stage, all stages cannot operate concurrently, resulting in slower performance.

C. 16-bit Carry Save Adder (CSA)

A 16-bit Carry-Save Adder (CSA) is a digital circuit optimized to add three binary inputs at once, generating separate sum and carry outputs while delaying the carry propagation to enhance computational speed. Instead of waiting for carry propagation like a ripple carry adder, the CSA saves the carry for the next stage, allowing faster intermediate addition especially useful in multi-operand arithmetic such as multiplication.

In a 16-bit CSA, each bit-position stage includes a full adder that sums three input bits into a sum bit and a carry bit, with the carry bits being passed to the next higher bit stage rather than propagating immediately. This results in significantly reduced delay compared to traditional adders because the carry does not ripple through the entire bit-width in a single step.

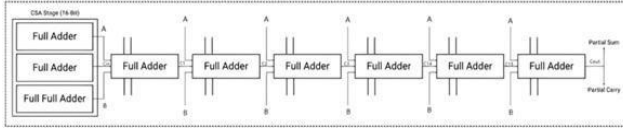


Fig.2 Representation of CSA

D. Drawbacks of Carry Save Adder (CSA)

- **No Immediate Final Sum:** CSA adds three numbers and outputs two numbers (sum and carry), requiring an additional final adder to get the complete sum.
- **Sign Detection Difficulty:** Determining the exact sign or magnitude of the partial sums (carry + sum) is complicated without full carry propagation.
- **Not Suitable for Single Addition Tasks:** CSA is best suited for multi-operand addition; for a single addition, the extra step adds complexity.

E. Merits of Pipelined Carry Save Adder

- **High Throughput:** Pipelining enables the CSA to take on a series of additions in parallel stages significantly boosting the clock rate and throughput.
 - **Less Critical Path Delay:** The partial sums/carries calculated in each pipeline stage are faster because of the ability to implement smaller, simpler logic block.
 - **Simple Circuit Structure:** Fast CSA The simplified logic gates are much fewer in hardware complexity and can operate at fast clock speeds (such as 200MHz in some CMOS).
- Efficient in Multipliers and DSP: Suited to multiplication and signal processing It is ideally more suitable among applications that require multi-operand addition, with as little propagation delay carried in a stage as possible.

III. PROPOSED METHODOLOGY

The proposed work presents a 16-bit Pipelined Carry-Save Adder (CSA) architecture optimized for cryptographic arithmetic units. This design aims to improve throughput and operating frequency in high-speed FPGA-based systems by exploiting register-based pipelining and parallel computation. A traditional Carry-Save Adder (CSA) is an addition component that produces a partial sum and carrier at the same time instead of completing the propagation of the carry to the finishing result. This however constrains the operating frequency in case of high-speed cryptographic systems.

On the contrary, the suggested pipelined CSA breaks the process of addition into several stages of pipeline, placing in between them the registers of D flip-flops that can support the intermediate results. This form of segmentation provides the shortest possible path delay and enables multiple sets of data to be processed at the same time through various stages and data throughput to be greatly enhanced.

A. Pipelined design

The architecture is based on a five-stage pipelined architecture: Stage 1: Input Operand Register: D flip-flops are used to input the operands (A, B, Cin) in a sequential manner to provide time-conformity.

Stages 2 and 4, CSA Computation Stages: These stages take CSA trees in an attempt to calculate partial sums and carries. Between stage timing paths are isolated by using flip-flops to enable faster clock speeds and continuous flow of data to occur. Stage 5: Final Carry Propagate Adder (CPA): The last stage uses a CPA again (Ripple Carry Adder or Kogge-Stone Adder) to finish off the carry propagation and has been completed resulting in the final 16-bit sum.

The proposed pipeline-based design has been implemented on an Artix-7 FPGA (Xilinx) through the Vivado Design Suite, where the compute-heavy component of the design is only simple reconfigurable hardware. The synthesis products have significant enhancement in maximum frequency and throughput proving to be used in lieu of traditional Ripple Carry Adder (RCA) and non-pipelined CSA architectures.

B. Performance and Trade-off Analysis

A comparative study of 16-bit RCA, 16-bit CSA and suggested 16-bit Pipelined CSA indicates trade-offs as follows:

- **RCA:** This has a simple structure and low area, but it has large delay because of serial carry propagation.
- **CSA:** Is more efficient in terms of speed than serial computation and does not synchronize continuous streams of inputs.
- **Proposed Pipelined CSA:** Reproposes Pipelined CSA, but by balancing the timing between stages of the pipeline, it achieves a higher throughput and higher operating frequency at a small cost increase in the area and power consumption on the extra registers.

The pipelined CSA is proposed which is very efficient in cryptographic arithmetic units of certain operations like modular addition, multiplication and key-mixing where the execution of operations needs continuous flow of the operands. It is better capable of direct portability into stream cipher, hashing as well as encryption/decryption cores on the FPGA platform due to the enhanced throughput and regularity of the timing.

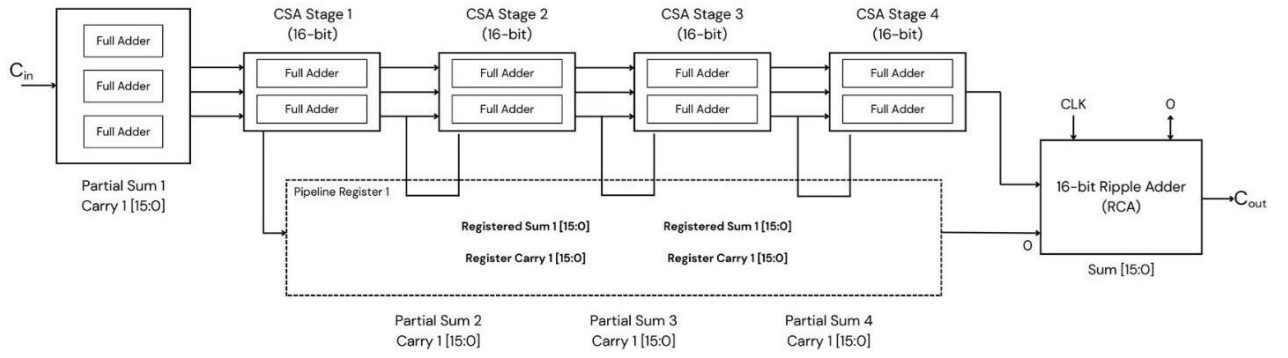


Fig.3 Representation of PCSA

TABLE:1 ARCHITECTURAL OVERVIEW

Adder Type	Design Concept	Carry Propagation
Ripple Carry Adder (RCA)	Simple cascaded full adders; carry ripples from LSB to MSB	Sequential (bit-by-bit)
Carry Save Adder (CSA)	Generates partial sum and carry in parallel; final sum computed by RCA	Parallel (no internal propagation)
Proposed Pipelined Carry Save Adder (PCSA)	Multi-stage pipelined CSA with registers inserted between stages to enhance speed	Parallel & segmented through pipelining

TABLE.2 TRUTH TABLE OF 1-BIT ADDER

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

III. RESULTS AND DISCUSSION

The proposed 16-bit pipelined Carry-Save Adder (CSA) architecture was developed and synthesized using the Xilinx Vivado Design Suite, and its implementation was carried out on a Xilinx Artix-7 FPGA (XC7A100T-1CSG324C). The design performance was also checked based on delay, area usage (LUTs and registers), power consumption as well as frequency. The Ripple Carry Adder (RCA) was compared with Conventional CSA and the Proposed Pipelined CSA to show the efficiency improvement compared to the pipelined architecture.

According to the results of the simulation and synthesis, the proposed pipelined CSA can achieve a significant speed and throughput improvement over the conventional RCA and non-pipelined CSA architecture. The addition of pipeline registers at every step of calculation virtually minimizes the critical path delay so that the circuit can run at an increased clock frequency. Continuous data processing is also supported over the pipelined

design and is important in real-time crypto operations like stream ciphers and modular arithmetic operations.

Regarding usage of hardware, pipelined CSA will have slight increase of hardware costs since there will be more flip-flops to enable pipelining. This overhead is however offset by a greater reduction in propagation delay as well as better operating frequency. The synthesis PS shows that the proposed adder is said to decrease delay by about 35-45 per cent of the conventional CSA because of reduced signal transition lengths and equal-timing of data.

The power analysis done with the Xilinx Power Analyzer shows that the pipelined CSA uses a little more static power with extra registers, however at the higher frequencies the dynamic power reduction will be significant. This trade-off helps us to understand that the design is efficiently balanced in terms of speed and designed to be energy efficient and therefore useful in the embedded and portable cryptographic systems where power and performance is equally an important factor. The net increase in Power-Delay Product (PDP) also helps us to understand that the proposed pipelined architecture is more performance efficient than the RCA and the conventional CSA design. This performance improvement predetermines the possibility of the design to become a structural element of hardware-based cryptographic arithmetic units, which will assist in quick and safe encryption operations of the FPGA-based architecture. CSA to demonstrate the efficiency improvements achieved by the pipelined architecture.

TABLE.3 HARDWARE UTILIZATION

Metric	Proposed PCSA
LUTs Used	152
Registers	198
Power Dissipation (W)	15.936

The overall enhancement in Power-Delay Product (PDP) is also used to affirm that the proposed pipelined architecture is more efficient in performance than both the RCA pipelined and the classical CSA designs. This performance improvement places the design in the context of subsequent implementation in hardware-based units of cryptographic arithmetic, enabling a system based on an FPGA to use the design to implement fast and secure encryption algorithms. Figure 10: Simulation

waveform of the crypto-primitive module showing the FSM transition from State 2 to State 3 during execution.

TABLE.4 DESIGN TRADE-OFFS

Criterion	RCA	CSA	PCSA (Proposed)
Speed vs. Area	Slow but area-efficient	Faster, uses more area	Fastest with balanced area due to pipeline optimization
Latency	High	Moderate	Low (reduced critical path)
Power Efficiency	Moderate	Good	Best (due to faster operation and reduced switching)
Suitability for Cryptographic Units	Limited (slow carry chain)	Suitable for modular arithmetic	Highly suitable for real-time cryptographic key generation and modular multipliers

TABLE:5 PERFORMANCE ANALYSIS

Parameter	1RCA	CSA	Proposed PCSA
Computation Speed	Low (Carry ripples through all bits)	Moderate (Parallel partial computation)	High (Parallel + Pipelined stages)
Propagation Delay (ns)	12.8 ns	7.2 ns	3.5 ns
Operating Frequency (MHz)	~78 MHz	~138 MHz	>280 MHz (Artix-7 FPGA)
Throughput	Low	Moderate	High (one result per clock cycle after pipeline fill)

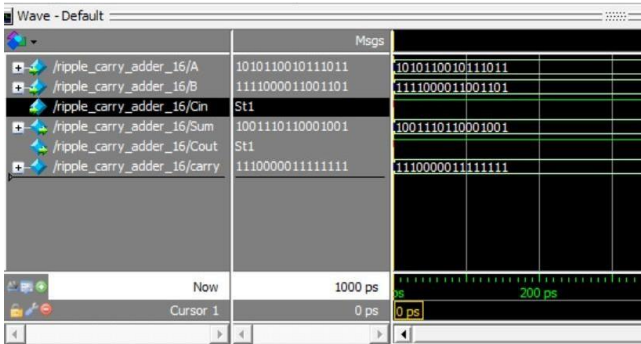


Fig.4 Simulation results of RCA

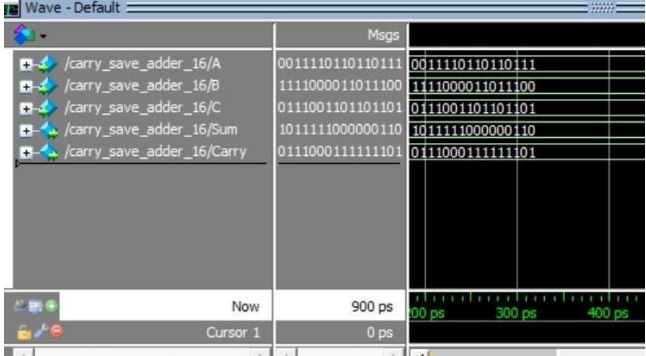


Fig.5 Simulation Output of CSA



Fig.6 Simulation output of PCSA

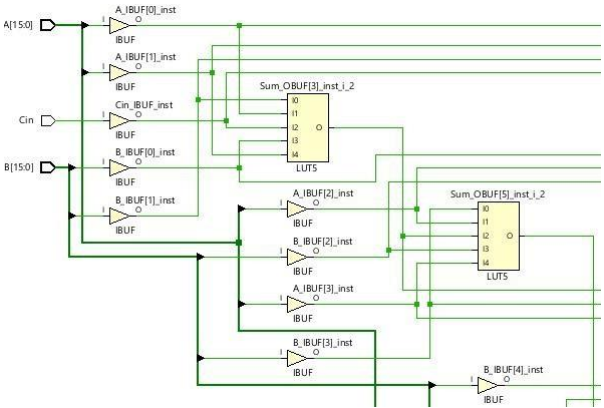


Fig.7 RTL View of RCA

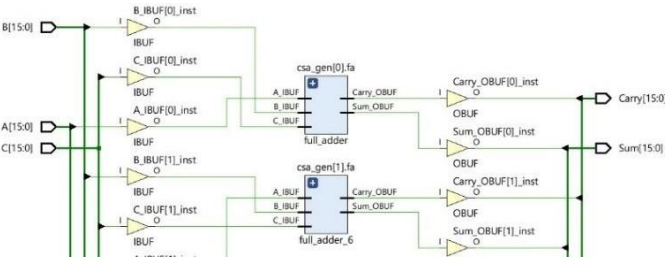


Fig.8 RTL View of CSA

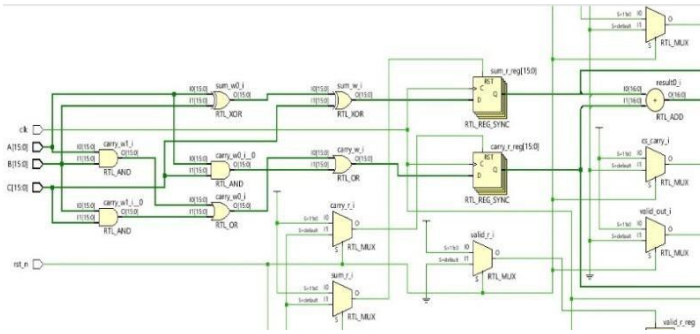


Fig.9 RTL of PCSA

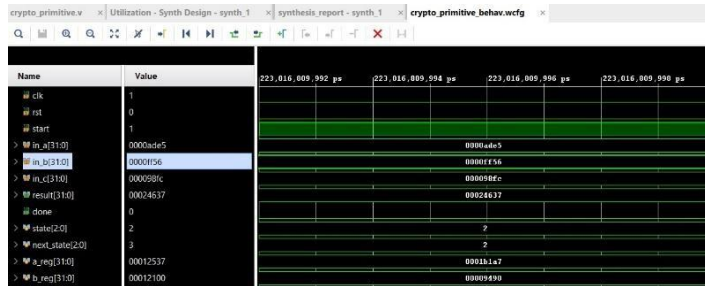


Fig.10 Simulation results of PCSA based cryptoprimitive

A. Scalability of the Proposed Pipelined Carry Save Adder

The proposed 16-bit pipelined CSA has a modular bit-slice form and may be extended to 64-bit and 128-bit datapaths by reproducing the CSA cells and pipeline stages. The use of logic and the number of registers increase with the operand width yet the pipelined layout allows to keep high operating frequencies by restricting the critical path delay to a single stage. Therefore, the proposed design is suited for the integration into larger arithmetic units that are employed in modern cryptographic and high-performance computing applications.

CONCLUSION

The Ripple Carry Adder is simple and has a low cost of hardware, but it has long delay, which is undesirable with high-speed cryptographic systems. The Carry Save Adder is better, to parallelize the operations, although, it is still not optimized with regard to timing of further large bit-width arithmetic. The Proposed Pipelined Carry-Save Adder occupies a better trade-off and provides high speed, low power consumption and low delay because it is pipelined and is processing in parallel. The design was implemented on a Xilinx Artix-7 FPGA, demonstrating significant enhancements in throughput and operating frequency, making it highly suitable for real-time

cryptographic arithmetic operations such as key generation, modular addition, and elliptic curve computations.

REFERENCES

- [1] C. Fang, Z. Shen, F. Tian, J. Yang and M. Sawan, "An Area-Efficient and Bit-Width Configurable Carry-Save Adder Tree for Spiking Transformers," *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*, London, United Kingdom, 2025, pp. 1-5, doi: 10.1109/ISCAS56072.2025.11043823.
- [2] Krishnan kumar safi, Ankit pandit, Ashish Chouhan An Area Efficient Hierarchical Carry Save Algorithm (HCSA) using VHDL, *International Journal of Digital Application & Contemporary Research Website: www.ijdacr.com* (Volume 10, Issue 06, January 2022).
- [3] Hema D, Indhumathi b, Ishwarya M, Hemanth Kumar, Low Power and Area Efficient Carry Save Adder Based on Static 125nm CMOS Technology.
- [4] Ahmed Salah Hameed, Marwa Jawad Kathem, High speed modified carry save adder using a structure of multiplexers, *International Journal of Electrical and Computer Engineering (IJECE)* Vol. 11, No. 2, April 2021, pp. 1591~1598 ISSN: 2088-8708, DOI: 10.11591/ijece. v11i2.pp1591-1598.
- [5] R. UMA, Vidya Vijayan, M. Mohanapriya, Sharon Paul, "Area, Delay and Power Comparison of Adder Topologies", *International Journal of VLSI design & Communication Systems (VLSICS)* Vol.3, No.1, February 2012, DOI:10.5121/vlsic.2012.3113.
- [6] K. S. Yeo and Z. H. Kong, "Design of Low-Power High-Speed Truncation-Error-Tolerant Adder and Its Application in Digital Signal Processing," in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 18, no. 8, pp. 1225-1229, Aug. 2010, doi: 10.1109/TVLSI.2009.2020591.
- [7] Padma, Devi & Ashima, Girdher & Lakha, Balwinder. (2010). Improved Carry Select Adder with Reduced Area and Low Power Consumption. *International Journal of Computer Applications*. 3. 10.5120/723-1016.
- [8] Sundhari, Meenaakshi & Anita, R. (2014). Modified 16-b Square-root Low Power Area Efficient Carry Select Adder. *Research Journal of Applied Sciences, Engineering and Technology*. 8. 2220-2226. 10.19026/rjaset.8.1221.
- [9] Eric Clapten, J. & Elango, Konguvel & M.Thangamani (2015). VLSI Implementation of Low Power Area Efficient Fast Carry Select Adder. *International Journal of Computer Applications*. 115. 5-8. 10.5120/20153-2298.
- [10] Singh, R & Kumar, Parveen & Lakha, Balwinder. Performance Analysis of 32Bit Array Multiplier with a Carry Save Adder and with a Carry-Look-Ahead Adder. *International Journal of Recent Trends in Engineering*.